Homework set 6

Martingale convergence, Doob's decomposition

Martingale Theory with Applications, 1st teaching block, 2026

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Problems with •'s are to be handed in. These are due in Blackboard before noon on Thursday, 19th November. Please show your work leading to the result, not only the result. Each problem worth the number of •'s you see right next to it. Make sure you find all 10 •'s!

Use of AI: Category 2: Minimal - for example, using spell and grammar checkers to help identify mistakes but not rewrite chunks of text.

6.1 •••• *Pólya urn.* At time $n = 0$, an urn contains $B_0 = 1$ blue and $R_0 = 1$ red balls. At each time $n > 0$ a ball is chosen uniformly at random from the urn and returned to the urn, together with a new ball of the same colour. We denote by B_n and R_n the number of blue, respectively, red balls in the urn after the n^{th} turn of this procedure. Notice that $B_n + R_n = n + 2$. Let

$$M_n := \frac{B_n}{B_n + R_n}$$

be the proportion of blue balls in the urn just after turn n.

(a) Show that M_n is a martingale w.r.t. the natural filtration of the process.

(b) Show that B_n is discrete uniform: $\mathbb{P}\{B_n = k\} = \frac{1}{n+1}$ for $1 \le k \le n + 1$.

(c) Show that $M_\infty := \lim_{n\to\infty} M_n$ exists a.s. What is its distribution?

(d) Let T be the time the first blue ball is drawn. Show that $T < \infty$ a.s. *Hint: show that the events* $\{T > n\}$ *are decreasing and find the limit of their probabilities.*

(e) Show that $\mathbb{E}\,\frac{1}{T+2} = \frac{1}{4}$.

6.2 ••• Let X_1, X_2 ... be independent random variables with

$$\mathbb{P}\{X_n = i\} = \begin{cases} \mathrm{e}^{-n}, & \text{if } i = 0, \\ 1 - 2\mathrm{e}^{-n}, & \text{if } i = 1, \\ \mathrm{e}^{-n}, & \text{if } i = 2, \end{cases}$$

and $M_n = \prod_{k=1}^n X_k$.

(a) Show that M_n is a martingale w.r.t. the natural filtration.

(b) Show that M_n has an almost sure limit M_∞ that is almost surely finite.

(c) Construct a random variable Z that bounds M_n for each n, and is of finite mean.

(d) Show that $\mathbb{E}\,M_\infty = 1$.

6.3 Let X_1, X_2, ... be strictly positive i.i.d. random variables such that $\mathbb{E}\,X_1 = 1$ and $\mathbb{P}\{X_1 = 1\} < 1$.

(a) Show that $M_n = \prod_{i=1}^n X_i$ is a martingale w.r.t. the natural filtration.

(b) Deduce that there exists a real valued random variable L such that $M_n \to L$ a.s. as $n \to \infty$.

(c) Show that $\mathbb{P}\{L = 0\} = 1$. *Hint: argue by contradiction and note that if* M_n, $M_{n+1} \in (a - \varepsilon,\, a + \varepsilon)$ *then* $X_{n+1} \in \left(\frac{a-\varepsilon}{a+\varepsilon},\, \frac{a+\varepsilon}{a-\varepsilon}\right)$.

(d) Use the Strong Law of Large Numbers to show that there exists $c \in \mathbb{R}$ such that $\frac{1}{n} \ln M_n \to c$ a.s. as $n \to \infty$. Use Jensen's inequality to show that $c < 0$.

6.4 Let $X_n \in [0, 1]$ be adapted to $\mathcal{F}_n$. Let α, $\beta > 0$ such that $\alpha + \beta = 1$ and suppose

$$\mathbb{P}\{X_{n+1} = \alpha + \beta X_n \,|\, \mathcal{F}_n\} = X_n, \quad \mathbb{P}\{X_{n+1} = \beta X_n \,|\, \mathcal{F}_n\} = 1 - X_n.$$

Show:

(a) $\mathbb{P}\{\lim_n X_n = 0 \text{ or } 1\} = 1$. *Hint: Use martingale convergence, and try to find an independent sequence U_n that generates X_n.*

(b) If $X_0 = \theta$ then $\mathbb{P}\{\lim_n X_n = 1\} = \theta$.

6.5 (*Following Edward Crane's ideas; hard:*) Given are countably infinitely many gamblers, called $i = 1, 2, \ldots$, who respectively start with a fixed $X_0^{(i)} > 0$ (real) amount of money. After each day n, they have a non-negative $X_n^{(i)} \geq 0$ wealth. A gambler i goes *bankrupt* when they run out of money, $X_n^{(i)} = 0$, in which case they will never play again. Gamblers who are not bankrupt are called *active*. Each day $n > 0$ when there are at least two active gamblers, the first (in their index i) and second active gamblers pair up, as well as the third and fourth active gambler pair up, also the fifth and sixth, etc. If there is an odd number of active gamblers then the one with the highest index will not play on day n.

Each paired up gamblers $i < j$ play the following game between each other: they bet the smaller of the two wealths they have, $\min(X_{n-1}^{(i)}, X_{n-1}^{(j)})$, on an independent and fair coinflip. If the coin comes up Head, gambler i pays this amount to gambler j, and if the coin comes up Tail then gambler j pays this amount to gambler i, this is how $X_n^{(i)}$ and $X_n^{(j)}$ are calculated. The game stops when there is at most one active gambler. Show that, a.s.:

- If the total amount in play is finite i.e., $\sum_i X_0^{(i)} < \infty$, then in the $n \to \infty$ time-limit exactly one gambler will have all the money.
- If the total amount in play is infinite i.e., $\sum_i X_0^{(i)} = \infty$, then each gambler eventually goes bankrupt.

6.6 Recall the ABRACADABRA problem, and the variable X_n being the total wealth of all gamblers in play after the n^{th} letter has been typed. Perform Doob's decomposition on X_n.

6.7 Let $\Gamma_0 = 0$, and Γ_1, Γ_2, ... be the marks of a rate λ homogeneous Poisson process on $\mathbb{R}^+$.

(a) Perform Doob's decomposition on Γ_n.

(b) Show that the martingale M you found above is $\mathcal{L}^2$. Find its brackets process $\langle M \rangle$ and the Doob decomposition of M^2.

6.8 Fix $\beta < \lambda$, and for the Poisson process of Problem 6.7 define $X_n := \mathrm{e}^{\beta \Gamma_n}$. Perform Doob's decomposition on this process. (*It is not going to be very nice.*)

6.9 ••• *Decomposing in a product sense.* Let X_n be an adapted process, and assume that it is bounded: $\sup_n X_n < K < \infty$ and bounded away from zero: $\inf_n X_n > \delta > 0$ for some non-random K and δ. Show that

(a) there is

– a martingale $(M_n)_{n \geq 0}$ with $M_0 = 1$,

– a predictable process $(B_n)_{n\geq 0} > 0$ with $B_0 = 1$

such that $X_n = X_0 \cdot M_n \cdot B_n$. This decomposition is almost everywhere unique in the sense that for any other pair $(\widehat{M}_n,\ \widehat{B}_n)_{n\geq 0}$ with the above properties we have $\mathbb{P}\{M_n = \widehat{M}_n \text{ and } B_n = \widehat{B}_n \text{ for all } n \geq 0\} = 1$.

(b) $(X_n)_{n\geq 0}$ is a submartingale if and only if $\mathbb{P}\{B_n \leq B_{n+1} \text{ for all } n \geq 0\} = 1$ in the above decomposition.